1.1.8. Matrix Multiplic… 33:44

Write a program to multiply two matrices A and B. The program should read their dimensions and elements, check whether matrix multiplication is possible, and display the resulting product matrix.

If the number of columns in matrix A is not equal to the number of rows in matrix B, the program should skip reading the elements of the second matrix and print "Invalid input".

Input Format:

- The first line contains two integers, m and n, representing the dimensions of matrix A.
- The next m lines each contain n space-separated integers, representing matrix A.
- The next line contains two integers, n and p, representing the dimensions of matrix B.
- The next n lines each contain p space-separated integers, representing matrix B.

Output Format:

- Print the resulting matrix after multiplication, where each row is displayed on a new line with space-separated integers.
- If multiplication is not possible (number of columns of A ≠ number of rows of B), print "Invalid input".

Sample Test Cases

Test case 1

Explorer

CTC77533.c

Submit

Debugger

```c
#include<stdio.h>

int main(){
    int a[100][100],b[100][100];
    int m,n,p,r;
    scanf("%d %d",&m,&n);
    int i,j;
//SHOURYAM GARG 25070521094
    for(i=0;i<m;i++){
        for(j=0;j<n;j++){

scanf("%d",&a[i][j]);
        }
    }
    scanf("%d %d",&r,&p);
    if(n!=r){
        printf("Invalid input\n");
        return 0;
    }
    for(i=0;i<r;i++){
        for(j=0;j<p;j++){

scanf("%d",&b[i][j]);
        }
        //SHOURYAM GARG 25070521094
    }
    int c[100][100]={0};

    for(i=0;i<m;i++){
        for(j=0;j<p;j++){
            for(int k=0;k<n;k++){
```

‹ Prev Reset Submit Next ›

```
1 2 3
4 5 6
3 2
7 8
9 10
11 12
58 64
139 154
```

Test case 2

```
1 2
12 45
1 2
Invalid input
```

```
                c[i]
[j]+=a[i][k]*b[k][j];
            }
        }
    }
//SHOURYAM GARG
25070521094
    for(i=0;i<m;i++){
        for(j=0;j<p;j++)
{
            printf("%d
",c[i][j]);
            if(j<p-1){

printf("");
            }
        }
        printf("\n");
    }
}
//SHOURYAM GARG
```